

Hands-on Lab: String Patterns, Sorting and Grouping in MySQL using phpMyAdmin

Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database Used in this Lab

The database used in this lab is an internal database. You will be working on a sample HR database. This HR database schema consists of 5 tables called **EMPLOYEES**, **JOB_HISTORY**, **JOBS**, **DEPARTMENTS** and **LOCATIONS**. Each table has a few rows of sample data. The following diagram shows the tables for the HR database:

SAMPLE HR DATABASE TABLES									
EMPLOYEES									
EMP_ID	LAST_NAME	FIRST_NAME	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEPT_ID	
E1001	John	Thomas	M	5613 Rex, Oak Park, IL	100	100000	30001	2	
E1002	Alice	James	F	980 Berry Ln, Elgin, IL	200	80000	30002	5	
E1003	Steve	Wells	M	271 Sempas, Graysville, TN	300	50000	30002	5	
JOB_HISTORY									
EMP_ID	START_DATE	JOB_ID	DEPT_ID		JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY	
E1001	2000-01-30	100	2		100	Sr Architect	60000	100000	
E1002	2010-08-16	200	5		200	Sr Software Developer	60000	80000	
E1003	2010-08-10	300	5		300	Jr Software Developer	40000	60000	
DEPARTMENTS									
DEPT_ID	DEPT_NAME	MANAGER_ID	LOC_ID		LOCATIONS				
2	Architect Group	30001	LO001		LOC_ID	LOC_NAME	LOC_ADDRESS	LOC_CITY	LOC_STATE
5	Software Development	30002	LO002		LO001			2	
7	Design Team	30003	LO003		LO002			5	
					LO003			7	

https://cf-courses-data.s3.us.cloud-object-storage.opepdomain.cloud/IBMDeveloperSkillsNetwork-DB0201EN-SkillsNetwork/Labs/MySQL/week2/Create_and%20Load.md.html

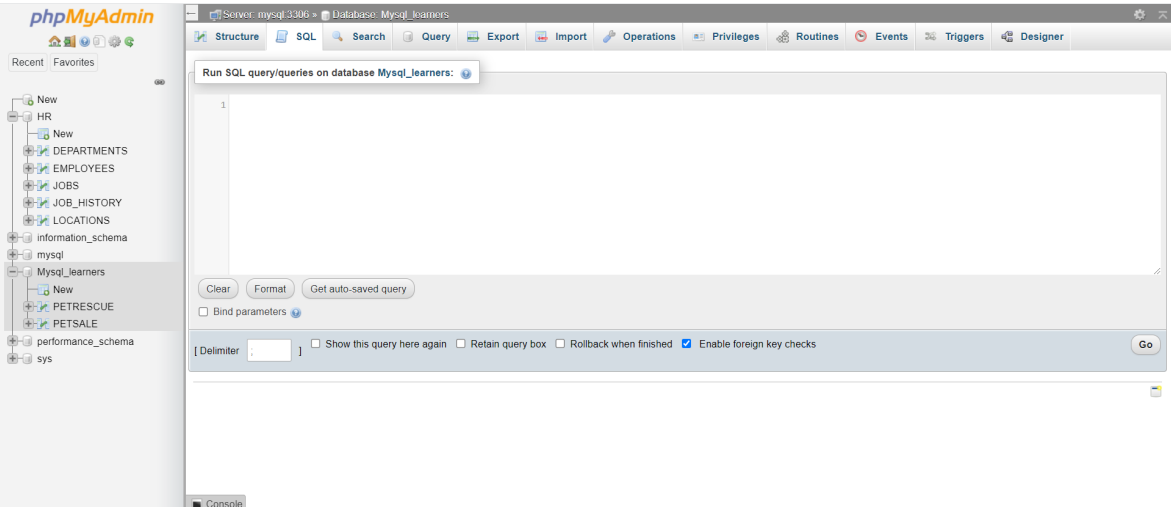
Once you have created the required tables, you may proceed with the exercises given in the lab.

Objectives

After completing this lab, you will be able to:

- Simplify a SELECT statement by using string patterns, ranges, or sets of values
- Sort the result set in either ascending or descending order and identify which column to use for the sorting order
- Eliminate duplicates from a result set and further restrict a result set

Once the tables are loaded open the sql editor to start executing the functions.



Exercise 1: String Patterns

In this exercise, you will go through some SQL problems on String Patterns.

1. Problem:
Retrieve all employees whose address is in Elgin, IL.

► Hint

Use the LIKE operator to find similar strings.

▼ Solution

```
SELECT F_NAME , L_NAME
FROM EMPLOYEES
WHERE ADDRESS LIKE '%1gIn,IL%';
```

▼ Output

Browse

Structure

SQL

Search

Insert

Ex

8

9 SELECT F_NAME , L_NAME

10 FROM EMPLOYEES

11 WHERE ADDRESS LIKE '%1gIn,IL%';

Extra options

← T →

▼ F_NAME L_NAME

☐

Edit

Copy

Delete

Alice

James

☐

Edit

Copy

Delete

Nancy

Allen

☐

Edit

Copy

Delete

Ann

Jacob

2. Problem:

Retrieve all employees who were born during the 1970's.

▼ Hint

Use the LIKE operator to find similar strings.

▼ Solution

```
SELECT F_NAME , L_NAME
FROM EMPLOYEES
WHERE B_DATE LIKE '197%';
```

▼ Output

SELECT F_NAME , L_NAME

FROM EMPLOYEES

WHERE B_DATE LIKE '197%';

← T →

▼ F_NAME L_NAME

☐

Edit

Copy

Delete

John

Thomas

☐

Edit

Copy

Delete

Alice

James

☐

Edit

Copy

Delete

Nancy

Allen

☐

Edit

Copy

Delete

Mary

Thomas

3. Problem:

Retrieve all employees in department 5 whose salary is between 60000 and 70000.

▼ Hint

Use the keyword BETWEEN for this SQL problem.

▼ Solution

```
SELECT *
FROM EMPLOYEES
WHERE (SALARY BETWEEN 60000 AND 70000) AND DEP_ID = 5;
```

▼ Output

2 of 6

1/6/2026, 10:45 PM

Server: phpMyAdmin demo - MySQLDatabase: HRTable: EMPLOYEES

BrowseStructureSQLSearchInsertExport

Run SQL query/queries on table HR.EMPLOYEES:

1 SELECT *

2 FROM EMPLOYEES

3 WHERE (SALARY BETWEEN 60000 AND 70000) AND DEP_ID = 5;

4

EMP_ID F_NAME L_NAME SSN B_DATE SEX ADDRESS JOB_ID SALARY MANAGER_ID DEP_ID

Edit Copy Delete E1004

Santosh Kumar

123456 1985-07-20 M

511 Aurora Av Aurora IL

400

60000.00

30004

5

Edit Copy Delete E1010

Ann Jacob

123415 1992-03-30 F

111 Entany Springs Egnr IL

220

70000.00

30004

5

Check allWith selectedEditCopyDeleteForward

Exercise 2: Sorting

In this exercise, you will go through some SQL problems on Sorting.

1. Problem:

Retrieve a list of employees ordered by department ID.

Hint

Use the ORDER BY clause for this SQL problem. By default, the ORDER BY clause sorts the records in ascending order.

Solution

SELECT F_NAME, L_NAME, DEP_ID
FROM EMPLOYEES
ORDER BY DEP_ID;

Output

1 SELECT F_NAME, L_NAME, DEP_ID

2 FROM EMPLOYEES

3 ORDER BY DEP_ID;

Edit Copy Delete

John Thomas 2

Edit Copy Delete

Ahmed Hussain 2

Edit Copy Delete

Nancy Allen 2

Edit Copy Delete

Alice James 5

Edit Copy Delete

Steve Wells 5

Edit Copy Delete

Santosh Kumar 5

Edit Copy Delete

Ann Jacob 5

Edit Copy Delete

Mary Thomas 7

Edit Copy Delete

Bharath Gupta 7

Edit Copy Delete

Andrea Jones 7

2. Problem:

Retrieve a list of employees ordered in descending order by department ID and within each department ordered alphabetically in descending order by last name.

Hint

Use the ORDER BY clause with DESC for this SQL problem.

Solution

SELECT F_NAME, L_NAME, DEP_ID
FROM EMPLOYEES
ORDER BY DEP_ID DESC, L_NAME DESC;

Output

3 of 6

1/6/2026, 10:45 PM

BrowseStructureSQLSearchInsertExportImport

Run SQL query/queries on table HR.EMPLOYEES:

```
1 SELECT F_NAME, L_NAME, DEP_ID
2 FROM EMPLOYEES
3 ORDER BY DEP_ID DESC, L_NAME DESC;
```

EditCopyDelete

MaryThomas7

EditCopyDelete

AndreaJones7

EditCopyDelete

BharathGupta7

EditCopyDelete

SteveWells5

EditCopyDelete

SantoshKumar5

EditCopyDelete

AliceJames5

EditCopyDelete

AnnJacob5

EditCopyDelete

JohnThomas2

EditCopyDelete

AhmedHussain2

EditCopyDelete

NancyAllen2

3. (Optional) Problem:

In SQL problem 2 (Exercise 2 Problem 2), use department name instead of department ID. Retrieve a list of employees ordered by department name, and within each department ordered alphabetically in descending order by last name.

► Hint

Department name is in the DEPARTMENTS table. So your query will need to retrieve data from more than one table. Don't worry if you are not able to figure this SQL problem out. We'll cover working with multiple tables in the lecture **Working with Multiple Tables**.

► Solution

```
SELECT D.DEP_NAME , E.F_NAME, E.L_NAME
FROM EMPLOYEES as E, DEPARTMENTS as D
WHERE E.DEP_ID = D.DEPT_ID_DEP
ORDER BY D.DEP_NAME, E.L_NAME DESC;
```

In the SQL Query above, D and E are aliases for the table names. Once you define an alias like D in your query, you can simply write D.COLUMN_NAME rather than the full form DEPARTMENTS.COLUMN_NAME.

► Output

```
1 SELECT D.DEP_NAME , E.F_NAME, E.L_NAME
2 FROM EMPLOYEES as E, DEPARTMENTS as D
3 WHERE E.DEP_ID = D.DEPT_ID_DEP
4 ORDER BY D.DEP_NAME , E.L_NAME DESC;
```

Extra options

DEP_NAME	F_NAME	L_NAME
Architect Group	John	Thomas
Architect Group	Ahmed	Hussain
Architect Group	Nancy	Allen
Design Team	Mary	Thomas
Design Team	Andrea	Jones
Design Team	Bharath	Gupta
Software Group	Steve	Wells
Software Group	Santosh	Kumar
Software Group	Alice	James
Software Group	Ann	Jacob

Exercise 3: Grouping

In this exercise, you will go through some SQL problems on Grouping.

NOTE: The SQL problems in this exercise involve usage of SQL Aggregate functions AVG and COUNT. COUNT has been covered earlier. AVG is a function that can be used to calculate the Average or Mean of all values of a specified column in the result set. For example, to retrieve the average salary for all employees in the EMPLOYEES table, issue the query: `SELECT AVG(SALARY) FROM EMPLOYEES;` You will learn more about AVG and other aggregate functions later in the lecture **Built-in Database Functions**.

1. Problem:

For each department ID retrieve the number of employees in the department.

► Hint

Use COUNT(*) to retrieve the total count of a column, and then GROUP BY.

► Solution

```
SELECT DEP_ID, COUNT(*)
FROM EMPLOYEES
GROUP BY DEP_ID;
```

► Output

```
1 SELECT DEP_ID, COUNT(*)
2 FROM EMPLOYEES
3 GROUP BY DEP_ID;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all

Number of rows: 25

Filter rows:

Extra options

DEP_ID	COUNT(*)
2	3
5	4
7	3

☐ Show all

Number of rows: 25

Filter rows:

2. Problem:

For each department retrieve the number of employees in the department, and the average employee salary in the department..

► Hint

Use COUNT(*) to retrieve the total count of a column, and AVG() function to compute average salaries, and then GROUP BY.

► Solution

```
SELECT DEP_ID, COUNT(*), AVG(SALARY)
FROM EMPLOYEES
GROUP BY DEP_ID;
```

► Output

```
1 SELECT DEP_ID, COUNT(*), AVG(SALARY)
2 FROM EMPLOYEES
3 GROUP BY DEP_ID;
```

Extra options

DEP_ID	COUNT(*)	AVG(SALARY)
2	3	86666.666667
5	4	65000.000000
7	3	66666.666667

3. Problem:

Label the computed columns in the result set of SQL problem 2 (Exercise 3 Problem 2) as NUM_EMPLOYEES and AVG_SALARY.

► Hint

Use SQL Aliases: column_name AS alias_name. For example, AVG(SALARY) AS "AVG_SALARY".

► Solution

```
SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"
FROM EMPLOYEES
GROUP BY DEP_ID;
```

► Output

```
1
2 SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"
3 FROM EMPLOYEES
4 GROUP BY DEP_ID;
```

☐ Show all

Number of rows: 25

Filter rows:

Extra options

DEP_ID	NUM_EMPLOYEES	AVG_SALARY
2	3	86666.666667
5	4	65000.000000
7	3	66666.666667

4. Problem:

In SQL problem 3 (Exercise 3 Problem 3), order the result set by Average Salary..

► Hint

Use ORDER BY after the GROUP BY.

► Solution

```
SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"
FROM EMPLOYEES
GROUP BY DEP_ID
ORDER BY AVG_SALARY;
```

► Output

```
1 SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"
2 FROM EMPLOYEES
3 GROUP BY DEP_ID
4 ORDER BY AVG_SALARY;
```

☐ Show all

Number of rows: 25

Filter rows:

Extra options

DEP_ID	NUM_EMPLOYEES	AVG_SALARY	▲ 1
5	4	65000.000000	
7	3	66666.666667	
2	3	86666.666667	

5. Problem:

In SQL problem 4 (Exercise 3 Problem 4), limit the result to departments with fewer than 4 employees.

- Hint
- Use HAVING after the GROUP BY, and use the count() function in the HAVING clause instead of the column label.
- Solution

```
SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"
FROM EMPLOYEES
GROUP BY DEP_ID
HAVING count(*) < 4
ORDER BY AVG_SALARY;
```

► Output

```
1
2 SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"
3 FROM EMPLOYEES
4 GROUP BY DEP_ID
5 HAVING count(*) < 4
6 ORDER BY AVG_SALARY;
```

☐ Show all

Number of rows: 25

Filter rows:

Extra options

DEP_ID	NUM_EMPLOYEES	AVG_SALARY	▲ 1
7	3	66666.666667	
2	3	86666.666667	

Solution Script

If you would like to run all the solution queries of the SQL problems of this lab with a script, download the script below.Import the script to phpadmin mysql interface and run. Follow [Hands-on Lab : Create tables using SQL scripts and Load data into tables](#) on how to upload a script to phpmyadmin console and run it.

- [StrinePattern-Sortine-Grooping_Solution_Script.sql](#)

Congratulations! You have completed this lab, and you are ready for the next topic.

Author(s)

[Lakshmi Holla](#)
[Malika Sinha](#)

